

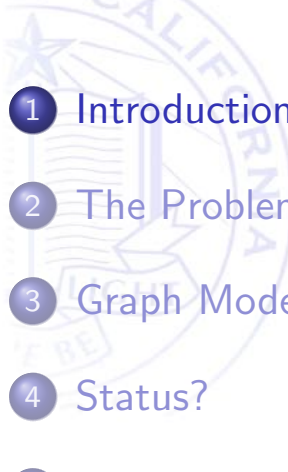


Graphical SLAM

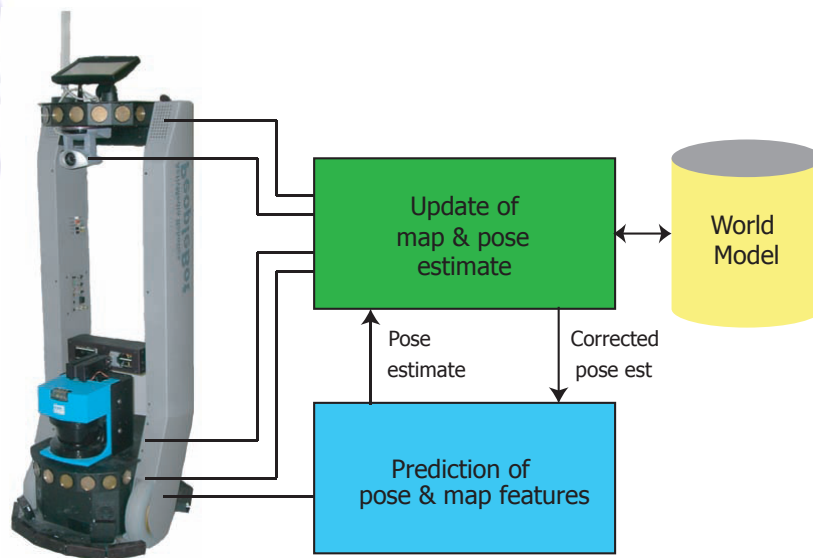
Henrik I. Christensen

Contextual Robotics
UC San Diego
La Jolla, CA
hichristensen@ucsd.edu

Outline

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- 1 Introduction
 - 2 The Problem
 - 3 Graph Model
 - 4 Status?
 - 5 Graph reduction
 - 6 Loop closing
 - 7 Example
 - 8 Summary

Outline of problem



Ensure the robot does not get lost!

The basics for SLAM

- State of robot is modelled as the pose

$$\vec{x}_R = \begin{bmatrix} x & y & \theta \end{bmatrix}^T$$

- Map features can be represented as points or lines, i.e.:

$$\vec{x}_i = \begin{bmatrix} x_i & y_i \end{bmatrix}^T$$

Estimation as a Kalman Problem

- Prediction by odometric modelling
- Updating as a Kalman process, with the state

$$\vec{x}_{state} = \begin{bmatrix} \vec{x}_R \\ \vec{x}_1 \\ \vdots \\ \vec{x}_n \end{bmatrix}$$

Why is SLAM difficult?

- The number of map hypotheses is very large
- Often the signal to noise ratio for features is ≈ 1
- Robust discriminative features are not common
- The “process” is often approximated

Problems?

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- ① Flexible inclusion/exclusion of measurements?
 - ② Handling of linearization?
 - ③ Dealing with topological constraints?
 - Loop closing etc.

Data handling

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- Easy inclusion and/or exclusion of data at any time in the process.
 - How to avoid too early a commitment to a particular map hypothesis.
 - Design of a representation that allow any-time inclusion/exclusion of data?

Linearizations?

- Linearization might cause divergence in the data.
- Reported by several, e.g. Julier & Uhlmann (2001); Castellanos *et al.* (2004)
- Consistent handling of non-linearities
 - Start by exact handling of non-linearities
 - As data matures a linearization is permitted
 - Identification of major non-linearities to include them

Topological constraints?

- Consistent inclusion of topological constraints
- Two step strategy:
 - 1 Close approximation of system in a trivial way
 - 2 Fine tune full model by adding “smaller” corrections

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Problem statement

- $\{x_i\}$ the robot path (set of poses), ($i \in \{1 \dots N_p\}$)
- $\{z_j\}$ feature coordinates ($j \in \{1..N_m\}$)
- $\{d_i\}$ dead reckoning measurements, between feature measurements
- $\{f_k\}$ feature measurements, ($k \in \{1..N_f\}$)
- Λ the $f \leftrightarrow z$ association

$$P(x, z, d, f, \Lambda) = P(d, f | x, z, \Lambda) P(x, z, \Lambda)$$


$$P(x, z, d, f, \Lambda) \propto P(d, f | x, z, \Lambda) P(x, z, \Lambda)$$

$$P(d, f | x, z, \Lambda) \propto P(d | x) P(f | x, z, \Lambda)$$

$$P(x, z, \Lambda) \propto P(\lambda) = P(N_f) \propto e^{-\lambda N_f}$$

$$P(x, z, d, f, \Lambda) \propto P(d | x) P(f | x, z, \Lambda) e^{-\lambda N_f}$$

An energy model

- 
- Definition of energy/entropy of the model:

$$E(x, z, d, f, \Lambda) = -\log(P(d | x)) - \log(P(f | x, z, \Lambda)) + \lambda N_f$$

- Or $E(x, z, d, f, \Lambda) = E_d + E_f + E_\Lambda$
- Or: ...

An energy model



$$E(x, z, d, f, \Lambda) = E_d(x) + E_f(x, z) + E_\Lambda(n_j) \quad (1)$$

$$E_d = - \sum_{i=1}^{N_p} \log(P(d_i | x_{i-1}, x_i)) = \frac{1}{2} \sum_{i=1}^{N_p} \xi_i^T k_i \xi_i \quad (2)$$

$$E_f = - \log(P(f | x, z, \Lambda)) = \frac{1}{2} \sum_{k=1}^{N_m} \eta_k^T k_k \eta_k \quad (3)$$

$$E_\Lambda = - \sum_{j=1}^{N_f} \lambda(n_j - 1) \quad (4)$$

$$\xi_i = T(x_i | x_{i-1}) - d_i \quad \eta_k = h(T(z_j | x_i)) - f_k$$

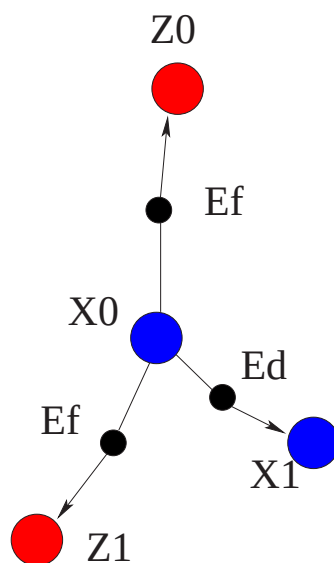
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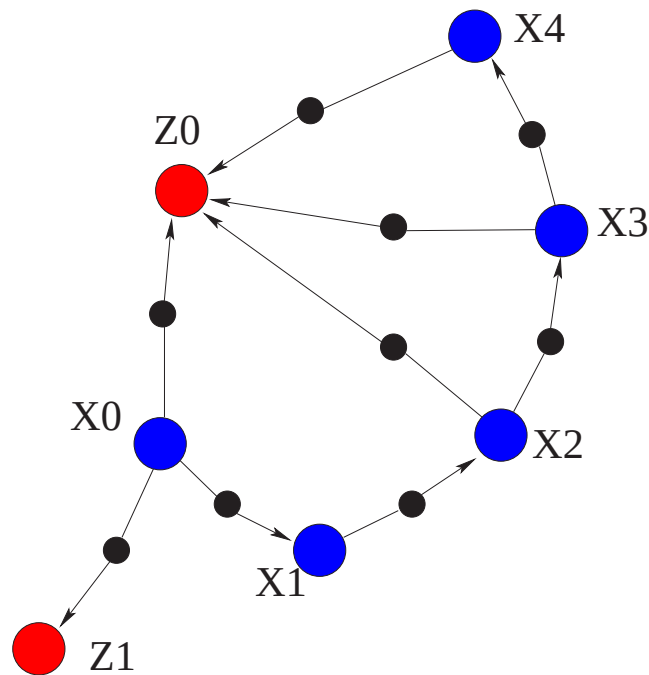
Organizing a model

- Graph representation
- Two types of nodes:
 - 1 State nodes
 - Poses (x_i)
 - Features (z_j)
 - 2 Energy Nodes (Computation of Eqn (1))
 - Connected to the state nodes needed for computation
 - Movement (d_i, k_i)

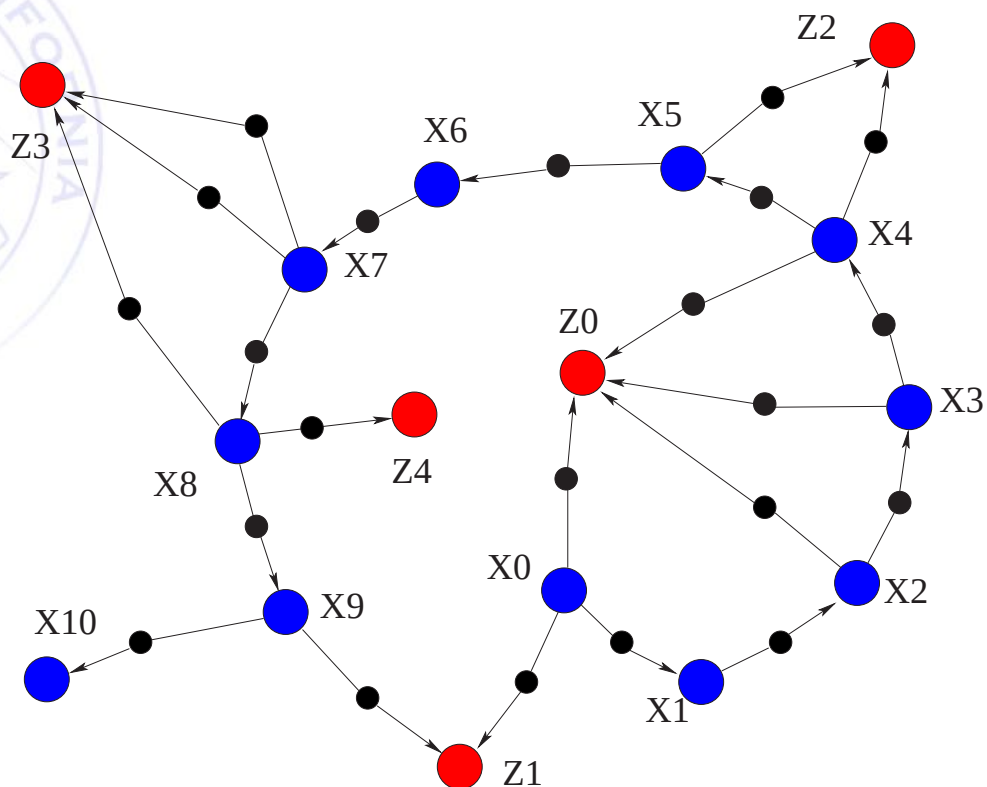
Starting a model



Entering more data



A graphical model example



- Optimal solution to Eq. (1): ($\text{argmin} E$) is not realistic.
- Relaxation techniques allow iterative updating
- In a time step:
 - 1 Add a new state node (pose)
 - 2 Any new features/measurements?
 - 3 Update the rest of the map – minimize energy

Map Update–Energy

- Eqn (1) Taylor expanded for a node A:

$$E_A = \sum_{i \in \text{edge}(A)} [E_i(\bar{x}_A, \bar{x}_i) + \nabla E_i(\bar{x}_A, \bar{x}_i) \begin{pmatrix} \Delta x_A \\ \Delta x_i \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \Delta x_A & \Delta x_i \end{pmatrix} \nabla^T \nabla E_i(\bar{x}_A, \bar{x}_i) \begin{pmatrix} \Delta x_A \\ \Delta x_i \end{pmatrix}]$$

Notation:

$$\mathcal{G} = \nabla E_A = \begin{pmatrix} \mathcal{G}_A \\ \mathcal{G}_i \end{pmatrix}$$

$$\mathcal{H} = \nabla^T \nabla E_A = \begin{pmatrix} \mathcal{H}_{AA} & \mathcal{H}_{Ai} \\ \mathcal{H}_{Ai} & \mathcal{H}_{ii} \end{pmatrix}$$

Map Update–Energy (II)

- Optimizing a node A:

$$(x_A - \bar{x}_A) = -\mathcal{H}_{AA}^{-1} \mathcal{G}_A \quad (5)$$

- With an energy change of

$$\Delta E_A = \frac{1}{2} \mathcal{G}_A^T \mathcal{H}_{AA}^{-1} \mathcal{G}_A \quad (6)$$

Map Update–Energy (III)

- ΔE_A decides on update strategy:
 - Steepest decent – close to saddle point
 - Use eqn. (5) direct
 - Chained update
 - Locate a node that is “good”
 - Update from that node

Feature Matching

- λ decides on the value of new measurements
 - 1 Add a new feature
 - 2 Compute energy change
 - 3 If change too large, remove association
- Similar to EM (Thrun *et al.* , 1998)
- Matching/graph updating anytime

Example

- ATRV Vehicle with Trimble DGPS 212
- Sick LMS 291 Laser scanner
- CrossBow DFOG INS package
- State $(\theta_n, x_n, y_n)^T$
- Lines w. end-points (x, y)
- See Folkesson & Christensen (2003) for details
- Operating around a house.
- 7500 Pose Nodes, 11975 line measurements
- Update time 30 ms (550 MHz Pentium)

SLAM example



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Status?

- Flexible association of data, anytime
- A framework to avoid linearization effects
- Loop closing:
 - 1 Recognition of loop closure (place recognition)
 - 2 Updating map to include the topological constraint
- Doing 2) often has limited effect
- A proposal ...

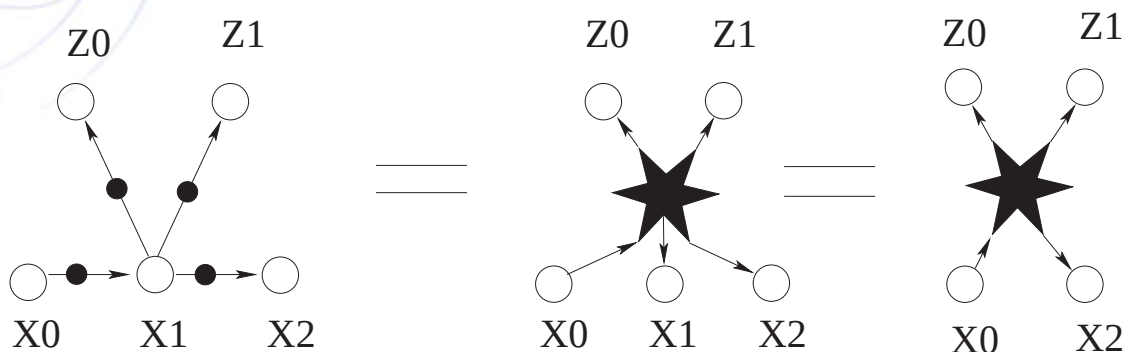
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Reducing the graph

- How can state nodes be removed from the graph?
- In a linear system one could generate a closed form solution
- In a non-linear case an approximation can be used.
- Consider elimination of a node A given neighboring nodes B.
- Consider the energy from state A expanded.

The basic idea



The Energy from A


$$E_A = \sum_{j \in \text{edge}(A)} [E_j(\bar{x}_A, \bar{x}_j) + \mathcal{G}_j(x_j - \bar{x}_j) + \frac{1}{2} \begin{pmatrix} x_A - \bar{x}_A & x_j - \bar{x}_j \end{pmatrix} \mathcal{H}_j(\bar{x}_A, \bar{x}_j) \begin{pmatrix} x_A - \bar{x}_A \\ x_j - \bar{x}_j \end{pmatrix}] \quad (7)$$

Transforming to B:

$$(x_A - \bar{x}_A) = -\mathcal{H}_{AA}^{-1} \mathcal{H}_{AB} (x_B - \bar{x}_B)$$

Elimination of A

- Substitution into eqn (7) eliminates A


$$E^* = \sum_{j \in \text{edge}(A)} [E_j(\bar{x}_A, \bar{x}_j) + \mathcal{G}_j(x_j - \bar{x}_j) + \frac{1}{2} (x_j - \bar{x}_j)^T \mathcal{H}_{jj} (x_j - \bar{x}_j) - \sum_{i \in \text{edge}(A)} (x_i - \bar{x}_i)^T \mathcal{H}_{iA} \mathcal{H}_{AA}^{-1} \mathcal{H}_{Aj} (x_j - \bar{x}_j)]$$

- New term connects i and j (new connection)

- The fused nodes are termed *Star* nodes.
- In principle the entire graph could be fused into a single node, as done in EKF.
- In practice star nodes are used to fuse local “maps”
- Typically upto 128 state nodes are considered for “fusion”.
- The star nodes are considered retroactively after say 50 poses.

Making star nodes invariant

- Star nodes updates energy using the Hessian $\mathcal{H}$
- If measurements have symmetries (such as lines) $\mathcal{H}$ will have zero eigenvalues.
- Project state nodes to “natural coordinates” (q) in a lower dimensional space without symmetries.

$$q_i = PT(x_i|x_0)$$

Mapping to natural coordinates

- In general:

$$\mathcal{H}_{xx} = J^T \mathcal{H}_{qq} J + \frac{\partial^2 q}{\partial x \partial x} \mathcal{G}_q \approx J^T \mathcal{H}_{qq} J$$

- Thus:

$$\mathcal{H}_{qq} = \tilde{J} \mathcal{H}_{xx} \tilde{J}^T$$

- Using SVD it is possible to find eigenvalues b_j and eigenvectors V_j of the Hessian

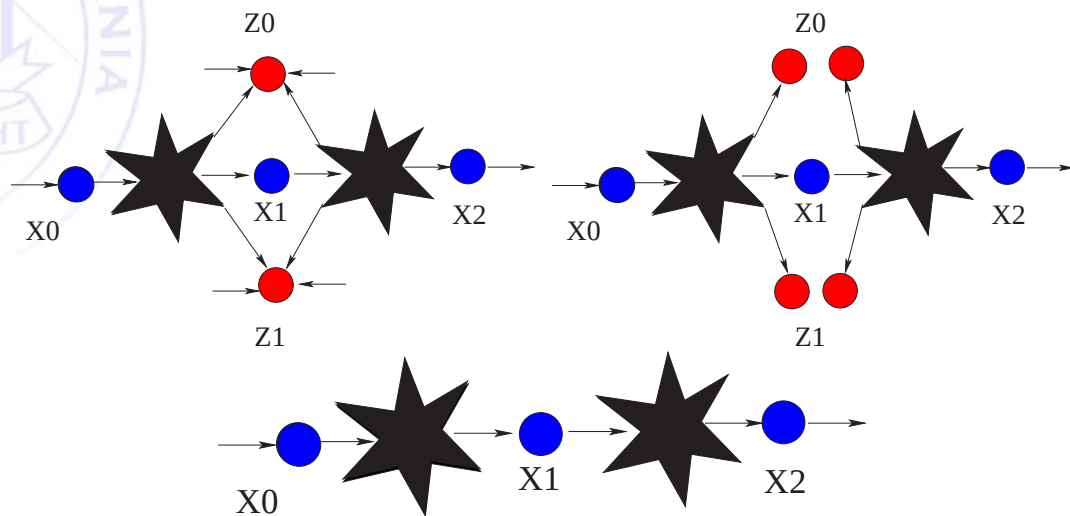
Energy & Properties

- Using the eigenvectors the energy at $\bar{q}$ is now:

$$E^* = E^*(\bar{q}) + \frac{1}{2} \sum_j b_j (V_j \Delta q)^2$$

- Recentered nodes are linear in energy
- A state node connected to only one star node can be eliminated ($\Delta q = 0$)
- Star nodes like local maps in Atlas (Bosse *et al.* , 2003)

Optimizing global calculations



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Closing loops

- With $\Delta q = 0$ the state nodes between star nodes is ignored.
- Define a cost function using Lagrange multipliers

$$C(\Lambda, \Delta q) = \Lambda \left(\sum_i \Delta x_i - d_c \right) + \frac{1}{2} \sum_i \sum_j b_{ji} (V_{ji} \Delta q)^2$$

- Where i is the star node index, Δx_i is the pose difference between the poses. d_c is the pose constraint.
- I.e.

$$\Delta x_i = \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} (\Delta q_i + \bar{q}_i)$$

Solving for closed loops

- Linearisation gives:

$$\Delta q_i = - \sum_j \frac{V_{ji}^T V_{ji}}{b_j} \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \Lambda^T$$

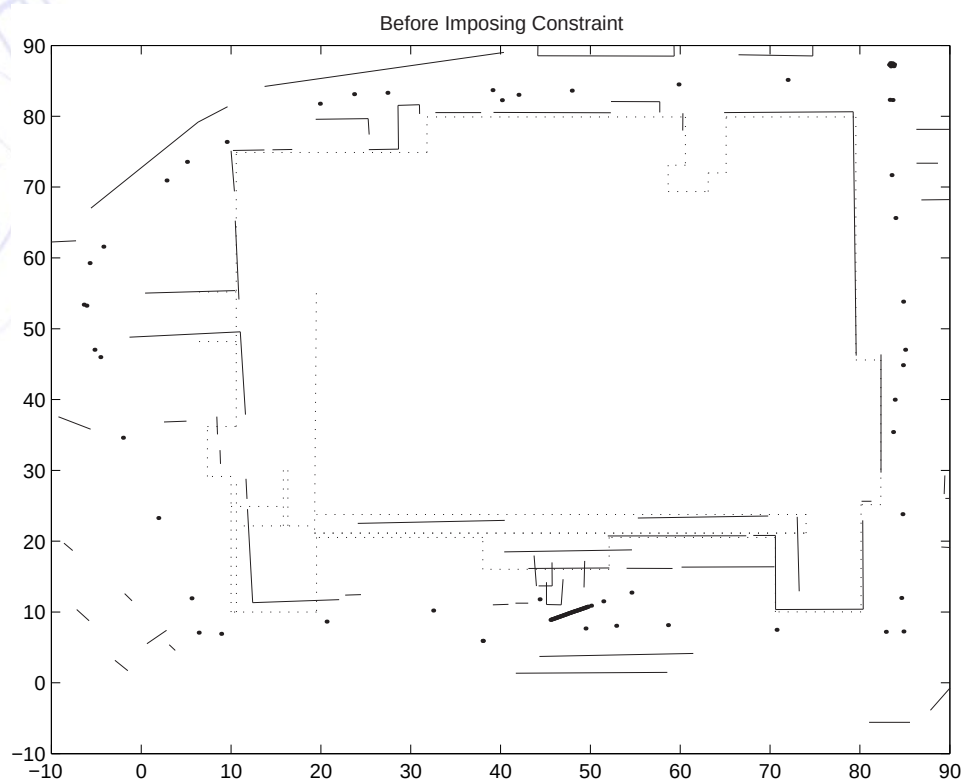
- Where:

$$\begin{aligned} \Lambda^T &= S^{-1} \left\{ \left[\sum_i \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \bar{q}_i \right] - d_c \right\} \\ S &= \sum_i \left\{ \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \left[\sum_j \frac{V_{ji}^T V_{ji}}{b_j} \right] \begin{pmatrix} \mathcal{R}_i & 0 \\ 0 & 1 \end{pmatrix} \right\} \end{aligned}$$

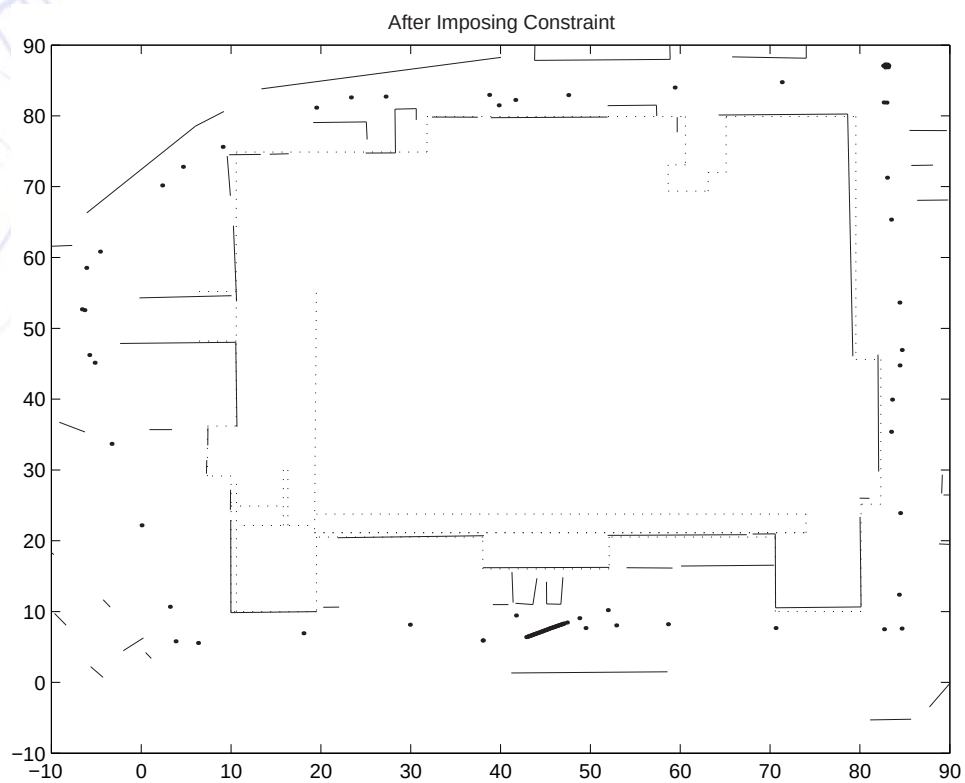
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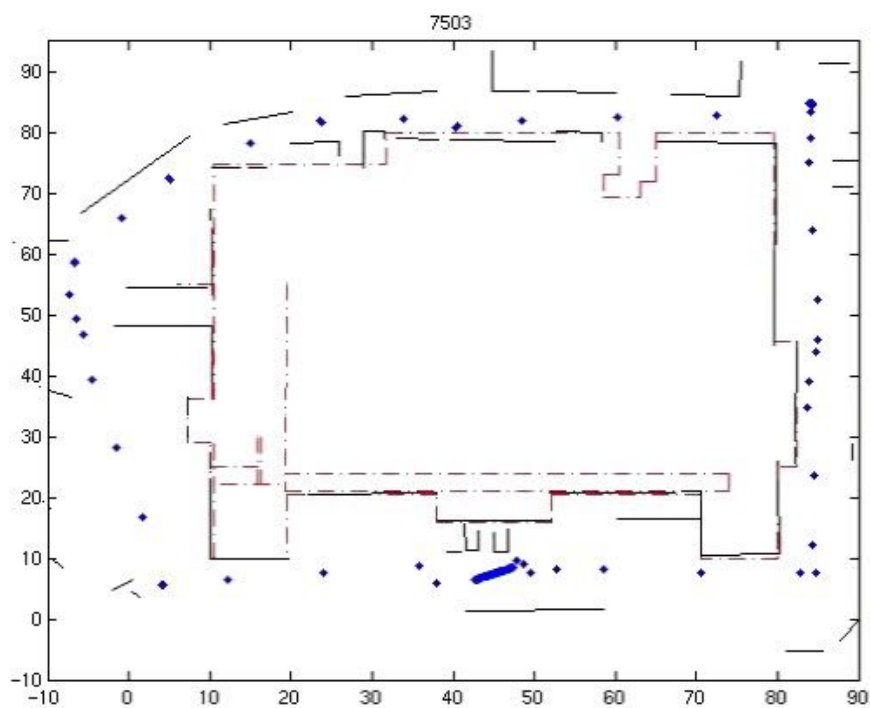
Map example - no closing



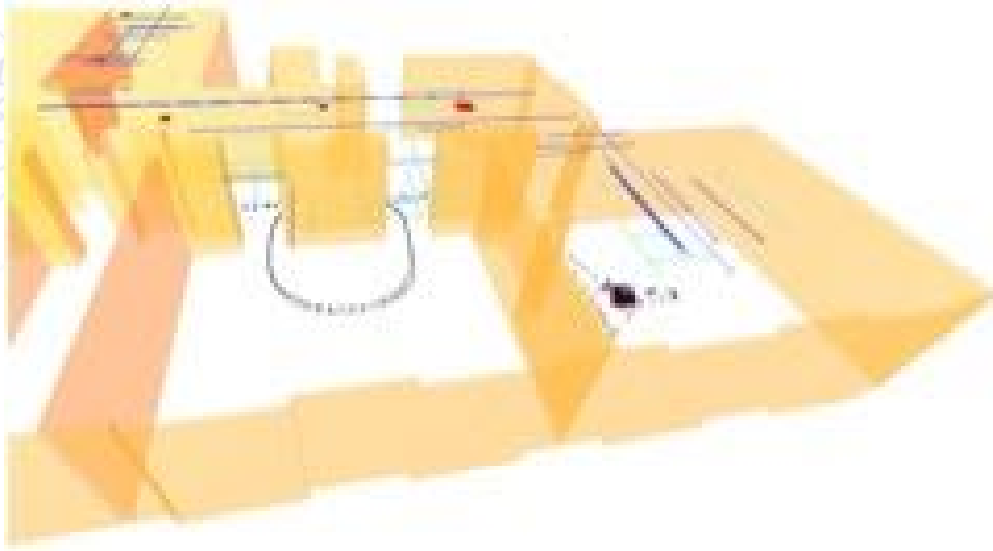
Map example - closing



Example Movie



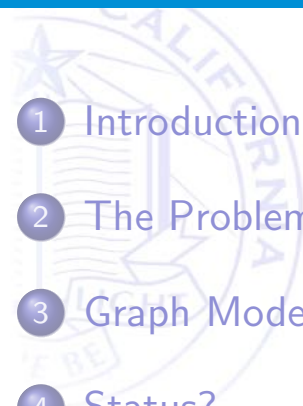
Example Movie 2



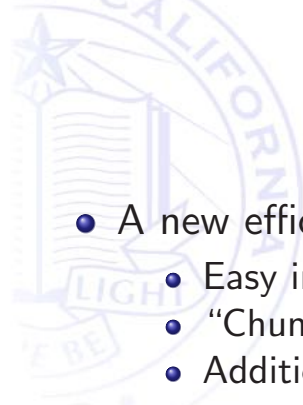
Closing loops

- Loop closing has a complexity of $O(N + 1)$ where N is the number of star nodes.
 - Once loop closing is achieved
 - Turn back on inter-star relation to fine tune
- Achieved in 1-2 seconds for large environments
- Constraints similar to “strong links” in Lu & Milios (1997).

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Summary

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- A new efficient representation for SLAM
 - Easy integration of data anytime
 - “Chunk-ing” data into local maps (star nodes)
 - Addition of constraint for loop closing
 - Computationally efficient (10-12 ms / pose)
 - Loop closing without linearization problems
 - Further details in Folkesson & Christensen (2004a,b); Fox *et al.* (2000); Thrun *et al.* (2005)

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